

React Beginner Tutorial

1. What is React?

React is a JavaScript library developed by Facebook for building user interfaces, especially single-page applications. It allows you to create reusable UI components.

2. Why React?

- Component-based architecture
- Fast rendering using Virtual DOM
- Easy to learn
- Large community support

3. Prerequisites

Before learning React, you should know:

- HTML
- CSS
- JavaScript (ES6 basics)

4. Installation

Install Node.js first, then run:

```
npx create-react-app myapp  
cd myapp  
npm start
```

5. Project Structure

src/ – main folder

App.js – root component

index.js – entry point

6. JSX

JSX allows writing HTML inside JavaScript.

Example:

```
const element = Hello React;
```

7. Components

Components are reusable UI blocks.

Function Component Example:

```
function Hello(){ return Hello World; }
```

8. Props

Props are used to pass data between components.

Example:

```
function Welcome(props){ return Hello {props.name}; }
```

9. State

State manages component data.

Example:

```
const [count,setCount] = useState(0);
```

10. Events

Click

11. Conditional Rendering

Render UI based on conditions.

12. Lists & Keys

Used to display data collections.

13. useEffect Hook

Used for side effects like API calls.

14. Forms

React uses controlled components.

15. React Router

Used for navigation between pages.

16. API Fetching

Use fetch or axios to get data from server.

17. Folder Best Practice

components/

pages/

services/

18. Build Project

npm run build

19. Example Mini App

Simple Counter App using useState.

20. Conclusion

React is powerful and beginner friendly. Practice daily and build projects.

Happy Learning React!